TireTwin: Modeling Tire Thermal Effects in Formula 1

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MATLAB/Simulink • FastF1 • Monte Carlo Analysis

Abstract

Tire thermal behavior is a major contributor to motorsport performance, but detailed tire-state measurements and proprietary tire models are not available in public Formula 1 telemetry. This study presents TireTwin, a MATLAB-based framework for investigating circuit-dependent thermal-performance sensitivity using public qualifying telemetry and a transparent transient thermal model. Telemetry from the 2024 Bahrain, Monaco, Monza, Silverstone, and Suzuka qualifying sessions was analyzed using the same driver (VER) across all circuits. A lumped thermal-state model estimates an effective tire thermal state from telemetry-derived operating-load proxies; a temperature-dependent friction formulation then maps thermal state to a simplified grip constraint and lap-time sensitivity. Under the baseline development parameterization, modeled lap-time penalties ranged from 1.430 s at Monaco to 2.366 s at Bahrain. Sensitivity analysis showed substantial dependence on heat-input scaling, effective thermal capacity, assumed optimum temperature, thermal-response width, and the imposed minimum-friction constraint. A 5,000-run Monte Carlo study (1,000 parameter realizations per circuit) produced strongly overlapping 95% simulation intervals: Bahrain had a median modeled penalty of 2.425 s (0.801-3.581 s), while Monaco had 1.616 s (0.427-3.178 s). The results therefore support TireTwin as an uncertainty-aware engineering analysis framework, not as a reconstruction of measured or proprietary Formula 1 tire temperatures. The central contribution is the explicit quantification of how simplified tire-model assumptions propagate into circuit-level performance conclusions.

Keywords: motorsport engineering; tire thermal modeling; Formula 1 telemetry; MATLAB; uncertainty quantification; sensitivity analysis; Monte Carlo simulation; lap-time modeling

1. Introduction

Race-car performance depends on coupled aerodynamic, powertrain, vehicle-dynamics, driver, environmental, and tire effects. Tire behavior is especially important because rubber friction and force capability vary with thermal state. Prior work has incorporated temperature into tire-force models, developed multi-node and thermo-mechanical tire models, and studied thermodynamic tire effects in racing optimization [1-7]. These studies also show why a single public-telemetry temperature estimate should not be confused with a physically measured tread, carcass, gas, or contact-patch temperature.

Public Formula 1 telemetry creates a different problem. FastF1 exposes lap timing and car telemetry channels such as speed, throttle, brake state, engine speed, gear, and DRS, but it does not provide the proprietary tire-state measurements required to calibrate a Formula 1 tire model [8]. TireTwin therefore asks a narrower and more defensible question: when a transparent simplified thermal-performance model is driven by real telemetry, how robust are the resulting circuit-level performance conclusions to uncertain model assumptions?

The framework was designed around three principles: (i) use real, reproducible telemetry inputs; (ii) keep the model equations and assumptions explicit; and (iii) test conclusions under parameter perturbation rather than treating one nominal simulation as ground truth. This makes the project a sensitivity and uncertainty study rather than a proprietary tire reconstruction.

2. Research Questions and Contributions

RQ1. How do telemetry-derived operating conditions produce different transient effective thermal-state trajectories across circuits under a common simplified tire model?

RQ2. How sensitive are modeled lap-time effects to assumptions concerning thermal capacity, heat generation, cooling, optimum operating temperature, thermal-response width, maximum friction scaling, and minimum available grip?

RQ3. Do circuit-level differences remain distinguishable when uncertainty in these model parameters is propagated through Monte Carlo simulation?

The contribution is not a new validated Formula 1 tire model. It is a reproducible workflow that connects public telemetry to a transparent thermal-state model, a normalized grip constraint, lap-time integration, one-at-a-time sensitivity experiments, a grip-floor diagnostic, and Monte Carlo uncertainty propagation.

3. Data and Experimental Design

3.1 Telemetry source and selection

Qualifying telemetry was extracted with FastF1 from five 2024 events. The same driver identifier, VER, was used across all circuits to reduce driver-identity variation. The extraction script selected the fastest accurate qualifying lap when available and exported Time, Distance, Speed, Throttle, Brake, RPM, Gear, and DRS. FastF1 documents these car telemetry channels and provides lap-level telemetry access [8].

Circuit	Driver	Lap no.	Selected lap	Samples
Bahrain	VER	16	01:29.179000	702

Monaco	VER	24	01:10.567000	536
Monza	VER	11	01:19.662000	610
Silverstone	VER	23	01:26.203000	659
Suzuka	VER	11	01:28.197000	685

The processed telemetry contains between 536 and 702 samples per selected lap. The five exported files are preserved through the telemetry manifest in the publication package.

3.2 Preprocessing

Telemetry is sorted by distance, non-finite distance/speed/time rows are removed, duplicate distances are removed, and speed is smoothed with a five-sample moving mean. Longitudinal acceleration is approximated numerically from speed and time. A lateral-load proxy is formed from the magnitude of speed multiplied by the spatial speed gradient. Brake is converted to a binary signal and throttle is constrained to 0-100%.

4. TireTwin Model

4.1 Telemetry-derived heat-load proxy

The heat-input signal is explicitly a proxy, not measured tire heat flow. It combines braking, positive longitudinal acceleration under throttle, a lateral-load proxy, and a speed-squared term:

$$\dot{Q}in = kbrake \cdot Brake \cdot v + kthrottle \cdot Throttle \cdot \max(ax,0) + klateral \cdot L + kspeed \cdot v^2$$

This formulation is intentionally transparent. Literature supports the broader physical premise that tire temperature evolves through energy generation, transfer, and dissipation, but the coefficients used here are development parameters rather than identified Formula 1 tire coefficients [2-5].

4.2 Lumped transient thermal state

$$mt\ cp\ dT/dt = \dot{Q}in - hA(T - Tamb) + ktrack(Ttrack - T)$$

The model represents one effective thermal state. It does not separate tread surface, tread bulk, carcass, inflation gas, belt, wheel, or contact-patch flash temperature. Higher-fidelity literature models explicitly resolve multiple thermal states and spatial effects [1-5].

4.3 Temperature-dependent grip

$$\mu(T) = \mu max\ \exp[-(T - Topt)^2 / (2\sigma T^2)]$$

A Gaussian temperature response provides a smooth optimum operating region, consistent with the general observation that tire friction and force capability depend strongly on temperature [1,5-7]. A lower bound is imposed as $\mu \geq fmin\ \mu max$ to prevent the simplified curve from collapsing toward zero. Because this bound can materially affect lap-time estimates, it is audited separately.

4.4 Performance constraint and lap time

The observed speed trace is reduced only in regions with higher lateral-load proxy. The normalized grip scale is $\sqrt{\mu/\mu_{max}}$, and the local reduction is weighted by the normalized lateral-load proxy. The thermally constrained speed trace is then integrated over distance:

$$t_{lap} = \int ds / v(s)$$

This is a sensitivity model rather than a full nonlinear tire-force, aerodynamic, or multibody vehicle-dynamics solver.

4.5 Baseline development parameters

Parameter	Baseline value	Status
Effective tire mass	10 kg	Development assumption
Effective heat capacity	1500 J/(kg·K)	Development assumption
Convective coefficient-area term hA	35 W/K	Development assumption
Track exchange coefficient	18 W/K	Development assumption
Ambient temperature	25 °C	Development assumption
Track temperature	40 °C	Development assumption
Initial effective thermal state	80 °C	Development assumption
Optimum temperature T_{opt}	95 °C	Development assumption
Thermal width σ_T	15 °C	Development assumption
Maximum friction scale μ_{max}	1.70	Normalized model scale
Minimum μ fraction	0.35	Model safeguard; separately audited

These values are not claimed to be proprietary or experimentally identified Formula 1 tire parameters. The literature review supports the model structure and the importance of thermal effects, not these exact nominal values. For that reason, the paper treats the nominal case as a baseline development parameterization and places the main evidential weight on sensitivity and uncertainty analysis.

5. Uncertainty and Sensitivity Methods

One-at-a-time perturbations were applied to effective heat capacity, convective cooling, track heat exchange, optimum temperature, thermal-response width, maximum friction scale, and global heat-input scaling. A separate grip-floor experiment varied the minimum friction fraction from 0.20 to 0.50.

Monte Carlo uncertainty propagation used 1,000 realizations per circuit (5,000 total). The random seed was fixed at 42 for reproducibility. Development uncertainty envelopes were applied around thermal capacity, cooling, track exchange, optimum temperature, thermal width, maximum friction scale, and a global heat-input multiplier. These distributions are exploratory uncertainty envelopes, not empirically fitted probability distributions. Monte Carlo sampling is a standard way to examine how uncertain inputs propagate through a model [9].

6. Results

6.1 Baseline cross-circuit results

Circuit	Penalty (s)	Mean state (°C)	Peak state (°C)	In window (%)	Mean μ
Bahrain	2.366	110.09	136.04	49.86	1.058
Monaco	1.430	100.74	117.39	78.36	1.315
Monza	1.678	112.11	140.73	45.41	1.019
Silverstone	1.845	109.39	135.09	51.75	1.061
Suzuka	1.704	111.36	136.33	46.57	1.024

Under the baseline parameterization, Bahrain produced the largest modeled lap-time penalty (2.366 s) and Monaco the smallest (1.430 s). Monza produced the highest peak modeled thermal state (140.73 °C), but not the largest lap-time penalty. This is an important structural result: within TireTwin, peak modeled temperature alone does not determine the performance effect. The location of thermal degradation relative to dynamically demanding portions of the lap also matters.

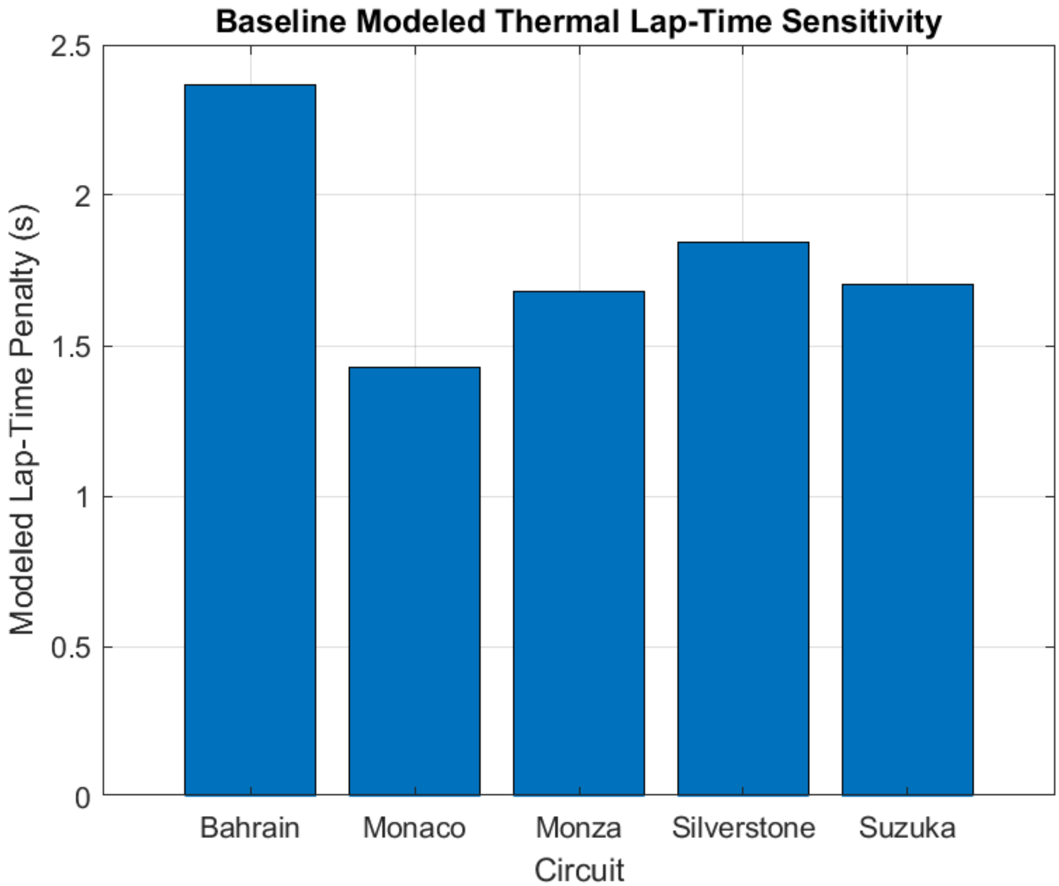

Figure 1. Baseline circuit-level thermal-performance sensitivity. Values are model outputs under the baseline development parameterization.

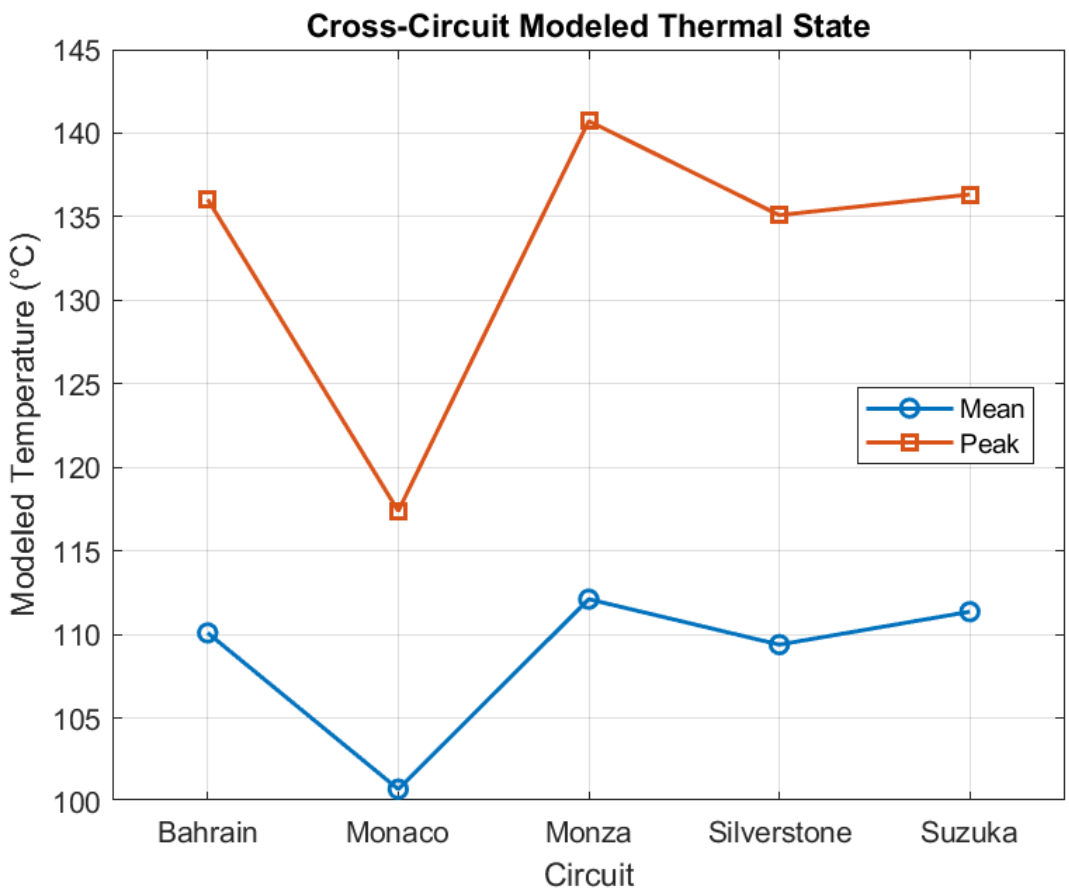

Figure 2. Mean and peak effective thermal states. These are modeled states, not measured Formula 1 tire temperatures.

6.2 Parameter sensitivity

Configuration	Mean modeled penalty (s)
Baseline	1.804
Heat input ×0.75	1.065
Higher thermal capacity	1.378
Wider thermal response	1.430
Lower thermal capacity	2.377
Heat input ×1.25	2.382
Topt -10 °C	2.588

The baseline mean modeled penalty across the five circuits was 1.804 s. Reducing global heat input by 25% lowered the mean penalty to 1.065 s, while increasing it by 25% raised the mean to 2.382 s. Lower effective thermal capacity increased the mean penalty to 2.377 s. Shifting the assumed optimum temperature downward by 10 °C produced the largest mean penalty among the reported perturbations, 2.588 s.

Changing μmax from 1.40 to 2.00 changed the reported absolute friction values but left lap-time penalties unchanged at every circuit. This is a structural invariance of the present normalized formulation: because the performance scale uses μ/μmax and μ(T) is proportional to μmax, the absolute scale cancels. Thus μmax is not identifiable from lap-time sensitivity in the current model.

6.3 Grip-floor diagnostic

Minimum μ fraction	Mean modeled penalty (s)
0.20	2.197
0.30	1.926
0.35	1.804
0.40	1.685
0.50	1.443

Raising the minimum friction fraction systematically reduced the absolute modeled penalty. The mean penalty changed from approximately 2.197 s at a 0.20 floor to 1.443 s at a 0.50 floor. Monaco was less affected than several other circuits at the lowest floor settings because its unconstrained minimum friction remained above the imposed 0.20 and 0.30 bounds. The diagnostic confirms that absolute lap-time penalties cannot be interpreted independently of this safeguard.

6.4 Monte Carlo uncertainty

Circuit	Median penalty (s)	2.5% (s)	97.5% (s)	Mean peak state (°C)
Bahrain	2.425	0.801	3.581	135.93
Monaco	1.616	0.427	3.178	117.90
Monza	1.706	0.656	2.640	141.84
Silverstone	1.952	0.924	2.725	136.17
Suzuka	1.748	0.611	2.423	137.27

The Monte Carlo analysis preserved a higher median penalty for Bahrain (2.425 s) than for the other circuits, but the 95% simulation intervals overlap strongly. Monaco, for example, spans 0.427-3.178 s, while Bahrain spans 0.801-3.581 s. These overlaps prevent a strong deterministic claim that one circuit intrinsically produces a larger thermal penalty than another under uncertain tire-model parameters.

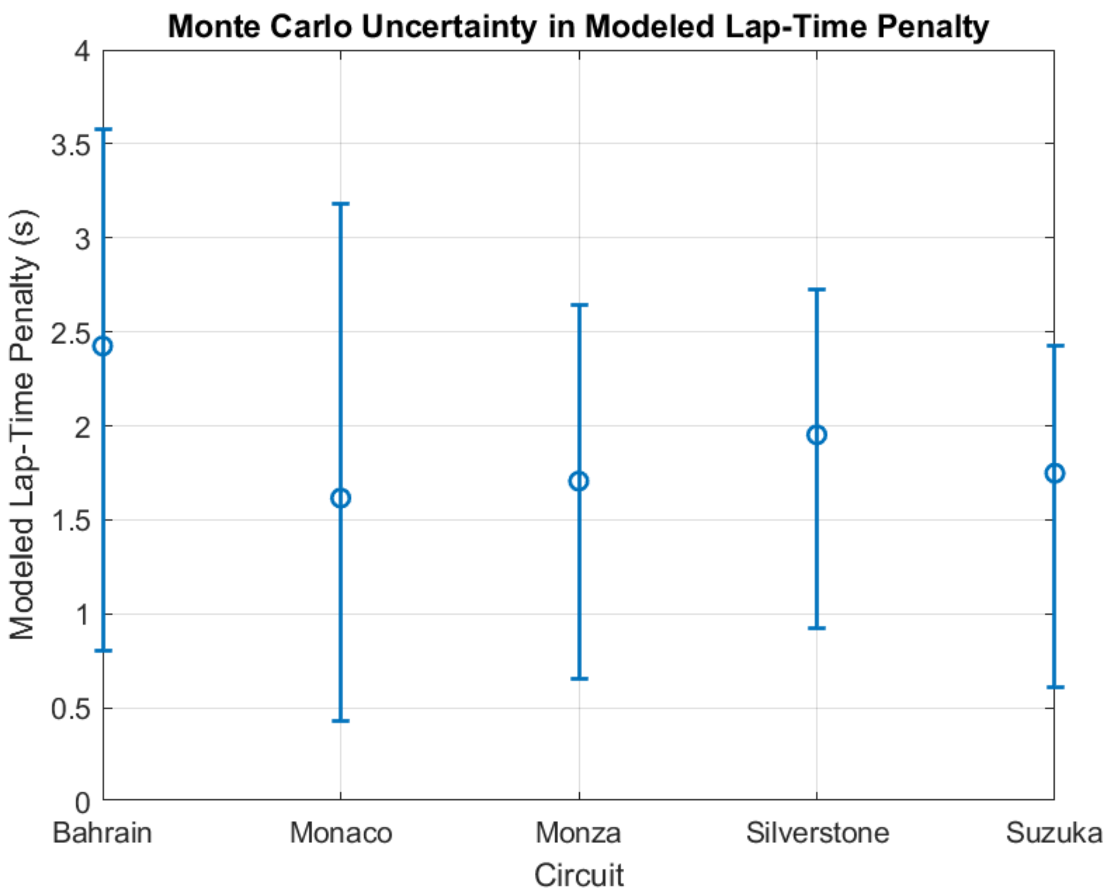

Figure 3. Median modeled lap-time penalty and 2.5th-97.5th percentile simulation intervals from 1,000 parameter realizations per circuit.

7. Discussion

The baseline experiment alone appears to provide a clear circuit ordering, but the uncertainty experiments substantially qualify that interpretation. Heat-input scaling, effective thermal capacity, optimum operating temperature, response width, and the minimum-friction safeguard materially alter absolute penalty estimates. This is the main research result: in a public-telemetry setting without proprietary tire-state measurements, model-form and parameter uncertainty are not secondary details; they directly govern the strength of the conclusions.

Monaco illustrates this point well. It has the greatest baseline thermal-window occupancy and the smallest baseline penalty, and it is less sensitive to the lowest grip-floor settings. Monza, by contrast, has the highest baseline peak effective thermal state but not the largest penalty. The model therefore distinguishes between a high thermal-state excursion and a performance-relevant thermal excursion.

The findings align with the broader tire-modeling literature in two ways. First, temperature is coupled to friction and force generation, so thermal state cannot generally be treated as irrelevant to performance [1,3,5-7]. Second, higher-fidelity models typically require multiple thermal states, experimentally identified parameters, and richer contact-patch physics [1-5]. TireTwin deliberately does not reproduce that fidelity. Its value is transparency: it makes the consequences of simplification measurable.

The invariance to μmax is also informative. It reveals a parameter that appears physically meaningful but has no effect on the modeled lap-time output because of the chosen normalization. Identifying such structural properties is useful before attempting calibration, because it prevents fitting parameters that the output cannot identify.

8. Limitations

- No measured tire temperature is available for validation; the reported thermal state must not be described as measured tread, carcass, surface, or gas temperature.
- The thermal model has one lumped state and omits spatial gradients, tread/carcass/gas separation, flash temperature, wheel/rim/brake coupling, pressure dynamics, and compound-specific behavior.
- Heat input is inferred from telemetry proxies. The coefficients are not measured heat-flow coefficients.
- Baseline parameter values and Monte Carlo distributions are development assumptions, not proprietary Formula 1 values or experimentally fitted distributions.
- The performance model is not a full tire-force, combined-slip, aerodynamic, suspension, or multibody vehicle model.
- Only one driver and one selected qualifying lap per circuit are used. This controls some variation but limits generalization across drivers, compounds, stints, fuel loads, and race conditions.
- Environmental quantities are fixed in the baseline model rather than reconstructed event-by-event.
- Monte Carlo sampling quantifies behavior under the chosen uncertainty envelopes; it does not guarantee worst-case bounds.

9. Reproducibility

The publication package includes the telemetry manifest, final baseline results, parameter-sensitivity results, grip-floor diagnostic, Monte Carlo summary, final figures, and the master MATLAB research-analysis script. The original starter project contains the model source functions and verification checks. The final run used a fixed Monte Carlo seed of 42. The exact MATLAB release used for the completed run was not captured in the provided artifacts; the project documentation recommends MATLAB R2024b or newer, with Simulink optional.

To reproduce the analysis, export the five qualifying telemetry CSVs with the required channels, place them in data/processed, run the model checks, and execute experiments/runResearchAnalysis.m. The final script performs 1,000 Monte Carlo realizations per circuit.

10. Conclusion

TireTwin demonstrates a complete telemetry-to-uncertainty workflow for motorsport engineering analysis. Real 2024 Formula 1 qualifying telemetry from Bahrain, Monaco, Monza, Silverstone, and Suzuka was coupled to a simplified transient thermal-state model, a temperature-dependent normalized grip model, and distance-domain lap-time integration. Baseline simulations produced circuit-dependent penalties from 1.430 s to 2.366 s, but sensitivity experiments showed that these absolute values depend materially on model assumptions. A 5,000-run Monte Carlo study further produced strongly overlapping circuit-level uncertainty intervals.

The defensible conclusion is therefore not that TireTwin reconstructs Formula 1 tire temperature or predicts an exact thermal lap-time loss. Rather, it shows how circuit-dependent thermal-performance patterns can emerge from real telemetry and how quickly those conclusions can change when uncertain tire-model assumptions are propagated. For public-data motorsport research, that uncertainty-aware framing is itself a useful engineering result.

Future work should incorporate literature-calibrated or experimentally identified parameter distributions, event-specific weather and track conditions, multiple drivers and compounds, stint-level evolution, multi-node thermal states, pressure and wear dynamics, and a higher-fidelity vehicle-performance model.

References

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Appendix A. Baseline Model Parameters

Parameter	Value
Effective tire mass	10 kg
Effective heat capacity	1500 J/(kg·K)
Convective coefficient-area term hA	35 W/K
Track exchange coefficient	18 W/K
Ambient temperature	25 °C
Track temperature	40 °C
Initial effective thermal state	80 °C
Optimum temperature Topt	95 °C
Thermal width σT	15 °C
Maximum friction scale μmax	1.70
Minimum μ fraction	0.35

Appendix B. Publication Integrity Statement

This artifact reports simulated quantities as simulated quantities. TireTwin does not claim access to proprietary Formula 1 tire models, internal team telemetry, measured tire temperatures, or confidential tire parameters. The nominal parameter set is a development baseline. Literature citations support the physical motivation and modeling context; they do not validate the exact nominal values used in this implementation.